MATH317 · NUMERICAL METHODS

# Midterm 461 Solutions

2024 · 60 minutes · 25 marks

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## 01 Solution Q1: errors and Newton

a) Absolute error = | exact − approximate |

Relative error = | exact − approximate | / | exact |

| x | Absolute error | Relative error |
|---|---|---|
| 4 | $\|2.0000 - 2.1000\| = 0.1$ | $0.1 / 2 = 0.05 = 5\ \%$ |
| 10 | $\|3.1623 - 3.1600\| = 0.0023$ | $0.0023 / 3.1623 = 0.073\ \%$ |

### Newton

$f'(x) = 3x^2 - 2$

$f(1.5) = -0.625$ $\quad$ $f'(1.5) = 4.75$

$x_1 = 1.5 - (-0.625)/4.75 = 1.6316$

$f(1.6316) = 0.0803$ $\quad$ $f'(1.6316) = 5.9864$

$x_2 = 1.6316 - 0.0803/5.9864 = 1.6182$

## 02 Solution Q2: Floating-point

Positive → s = 0

1. $(1259)_{10} = (2353)_8 = (100\ 1110\ 1011)_2$

2. $(.125)_{10} = (.1)_8 = (.001)_2$

3. $(1259.125)_{10} = (10011101011.001)_2 = (1.0011101011001)_2 \times 2^{10}$

4. c = 10 + 127 = 137 = $(211)_8$ = $(10001001)_2$

5. $[0 \mid 10001001 \mid 00111010110010000000000]_2$

   $[0100\ 0100\ 1001\ 1101\ 0110\ 0100\ 0000\ 0000]_2$

   $= [449D6400]_{16}$

The number is positive, so the first hex digit is below 8.

## 03 Solution Q3: Bisection

a) u = f(0) = 1 v = f(1) = e − 4 = −1.2817

   u · v < 0 → f has a root in [0, 1]

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b)  n = 0:  c = 0.5        w = f(0.5) = 0.6487

            u · w > 0   →   a = c   →   [0.5, 1]

    n = 1:  c = 0.75       w = f(0.75) = −0.1330

            u · w < 0   →   b = c   →   [0.5, 0.75]

    n = 2:  c = 0.625      w = f(0.625) = 0.3057

            u · w > 0   →   a = c   →   [0.625, 0.75]
```

| n | $c_n$ | $f(c_n)$ | Error |
|---|---|---|---|
| 0 | 0.5 | 0.6487 | 0.5 |
| 1 | 0.75 | −0.1330 | 0.25 |
| 2 | 0.625 | 0.3057 | 0.125 |